YEAR 9 MATHS · CHAPTER 4 · MIND MAP

Linear relationships

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1 Introduction to linear relationships

A linear rule gives a straight line. y changes by the same amount each step.

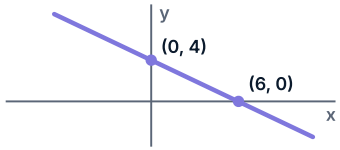
$y = 2x + 1$: $x = 0, 1, 2$ gives $y = 1, 3, 5$

Is $(3, 7)$ on the line? $2(3) + 1 = 7$ ✓ yes

2 Graphing using intercepts

x-intercept: put $y = 0$. y-intercept: put $x = 0$.

$2x + 3y = 12$: x-intercept $(6, 0)$, y-intercept $(0, 4)$



3 Lines with one intercept

Some lines only cross one axis, or pass through the origin.

$y = 3$: horizontal $x = -2$: vertical

$y = 2x$: through the origin $(0, 0)$

4 Gradient

Gradient $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$(1, 2)$ to $(3, 8)$: $m = \dfrac{6}{2} = 3$



Watch out: rise over run, not run over rise.

5 Gradient and direct proportion

$y = kx$: a straight line through the origin. The gradient k is the constant of proportion.

Paid $25 an hour: $P = 25h$

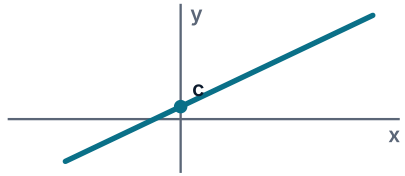
6 hours: $P = 25 \times 6 = \$150$

6 Gradient–intercept form

$y = mx + c$: m is the gradient, c is the y-intercept.

$y = -2x + 5$: $m = -2$, $c = 5$

Rearrange $3x + y = 4$: $y = -3x + 4$



7 Finding the equation of a line

Find m, then substitute a point to find c.

Gradient 2 through $(1, 5)$: $5 = 2(1) + c$, so $y = 2x + 3$

Through $(0, 1)$ and $(2, 7)$: $m = 3$, $c = 1$, so $y = 3x + 1$

8 Midpoint and length

Midpoint: average the coordinates. Length: Pythagoras.

$(2, 3)$ and $(8, 11)$: midpoint $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$

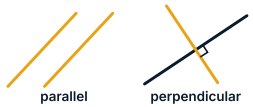
Length $\sqrt{6^2 + 8^2} = 10$

9 Parallel and perpendicular lines

Parallel: same gradient. Perpendicular: gradients multiply to -1.

$y = 2x + 1$ is parallel to $y = 2x - 4$

Perpendicular to $y = 2x + 1$: gradient $-\frac{1}{2}$



10 Linear modelling

c is the starting value, m is the rate of change.

Taxi: $4 flag fall plus $2.50 per km: $C = 2.5d + 4$

10 km: $C = 2.5 \times 10 + 4 = \$29$

11 Graphical solutions to simultaneous equations

The solution is the point where the two lines cross.

$y = x + 1$ and $y = -x + 5$ cross at $(2, 3)$

Check: $2 + 1 = 3$ ✓ and $-2 + 5 = 3$ ✓

Parallel lines never cross, so there's no solution.

